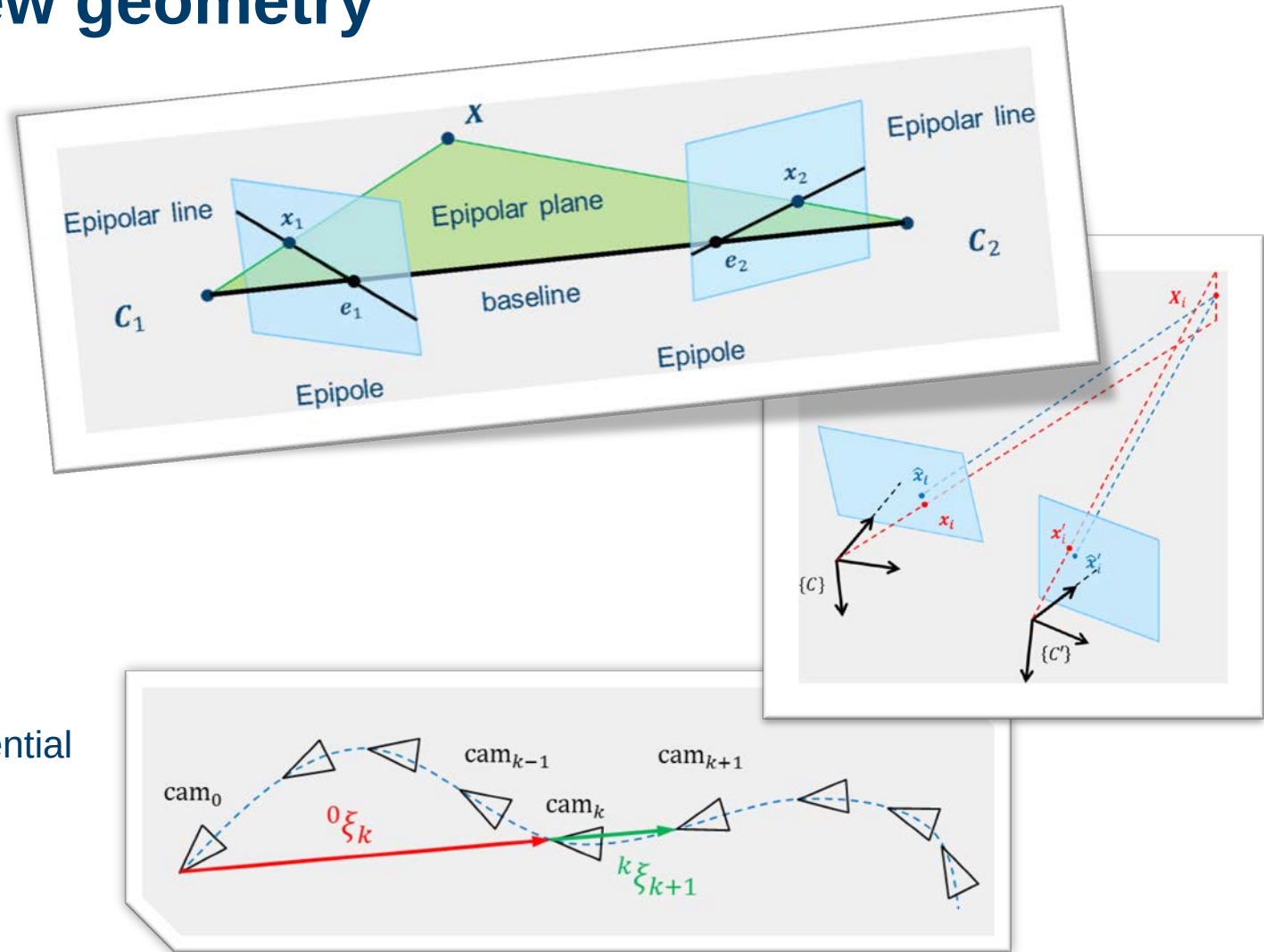


Lecture 7.1

Epipolar geometry

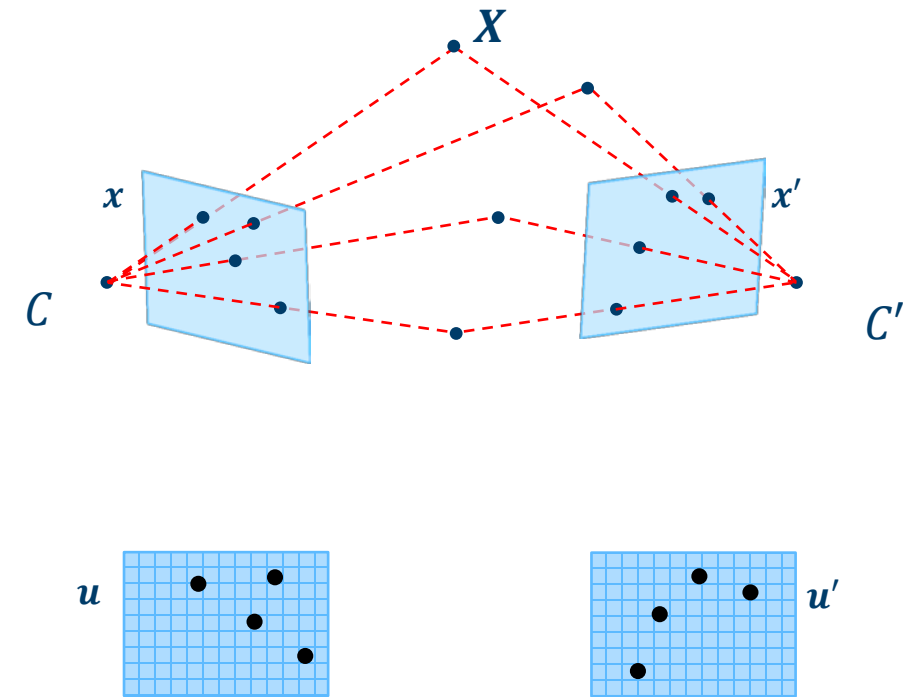
Weekly overview – Two-view geometry

- Epipolar geometry
 - Algebraic representation & estimation
 - The essential matrix
 - The fundamental matrix
- Triangulation
 - Sparse 3D scene reconstruction from 2D correspondences
- Relative pose from epipolar geometry
 - Estimating the relative pose from the essential matrix
 - Visual odometry



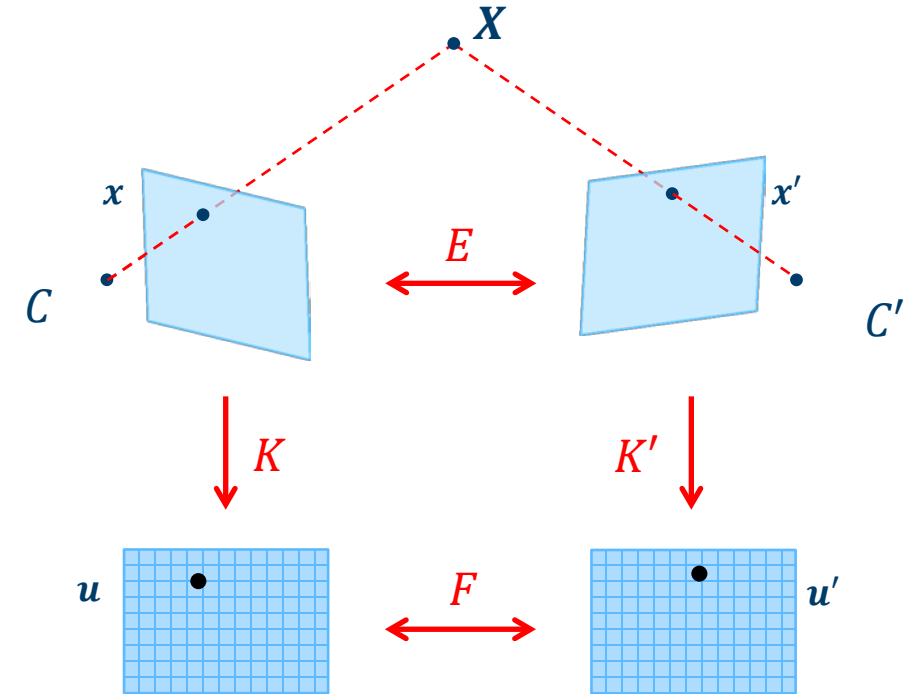
Introduction

- Observing the same points in two views puts a strong geometrical constraint on the cameras
- Algebraically this epipolar constraint is usually represented by two related 3×3 matrices



Introduction

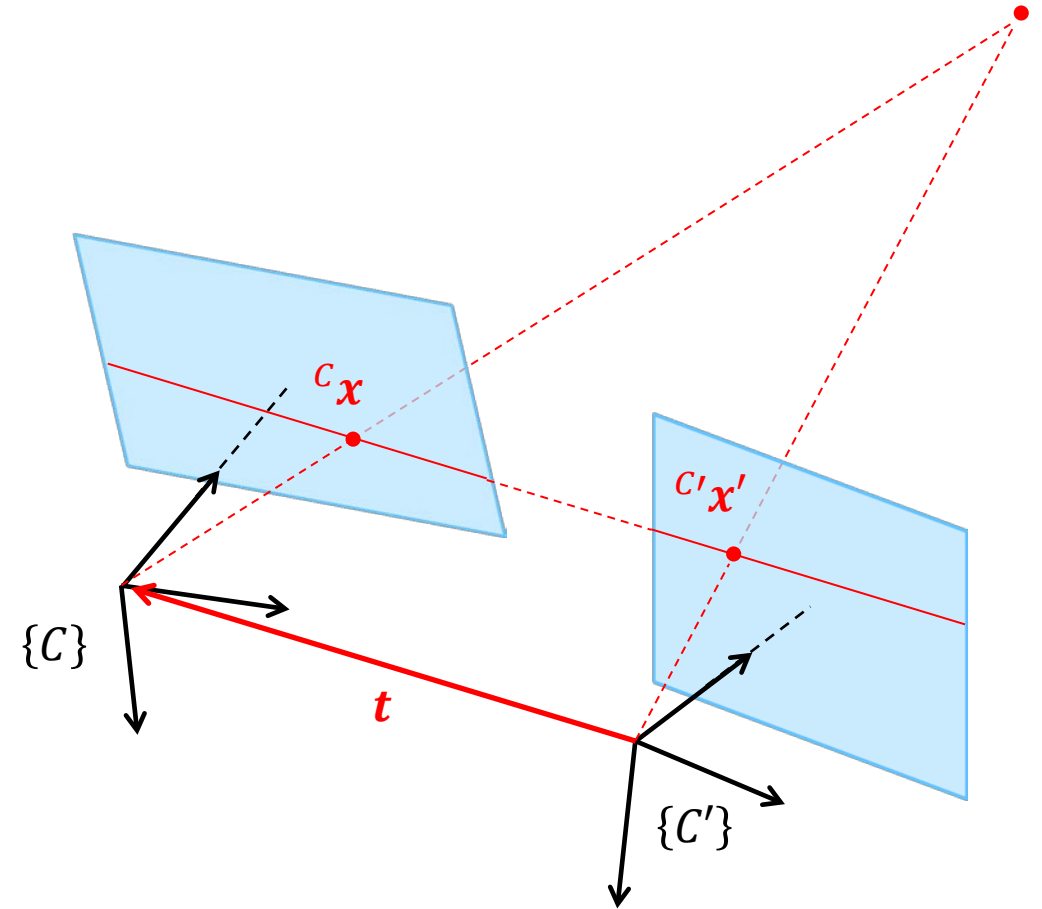
- Observing the same points in two views puts a strong geometrical constraint on the cameras
- Algebraically this epipolar constraint is usually represented by two related 3×3 matrices
- The fundamental matrix F
$$\tilde{\mathbf{u}}'^T F \tilde{\mathbf{u}} = 0$$
- The essential matrix E
$$\tilde{\mathbf{x}}'^T E \tilde{\mathbf{x}} = 0$$
- These are related through the two calibration matrices K and K'



The essential matrix E

- Let ${}^C\mathbf{x} \leftrightarrow {}^{C'}\mathbf{x}'$ be corresponding points in the normalized image planes and let the pose of $\{C\}$ relative to $\{C'\}$ be

$${}^{C'}\xi_C = \begin{bmatrix} R & \mathbf{t} \\ \mathbf{0} & 1 \end{bmatrix}$$



The essential matrix E

- Let ${}^C\mathbf{x} \leftrightarrow {}^{C'}\mathbf{x}'$ be corresponding points in the normalized image planes and let the pose of $\{C\}$ relative to $\{C'\}$ be

$${}^{C'}\xi_C = \begin{bmatrix} R & \mathbf{t} \\ \mathbf{0} & 1 \end{bmatrix}$$

- In terms of vectors, the equation for the epipolar plane can be written like

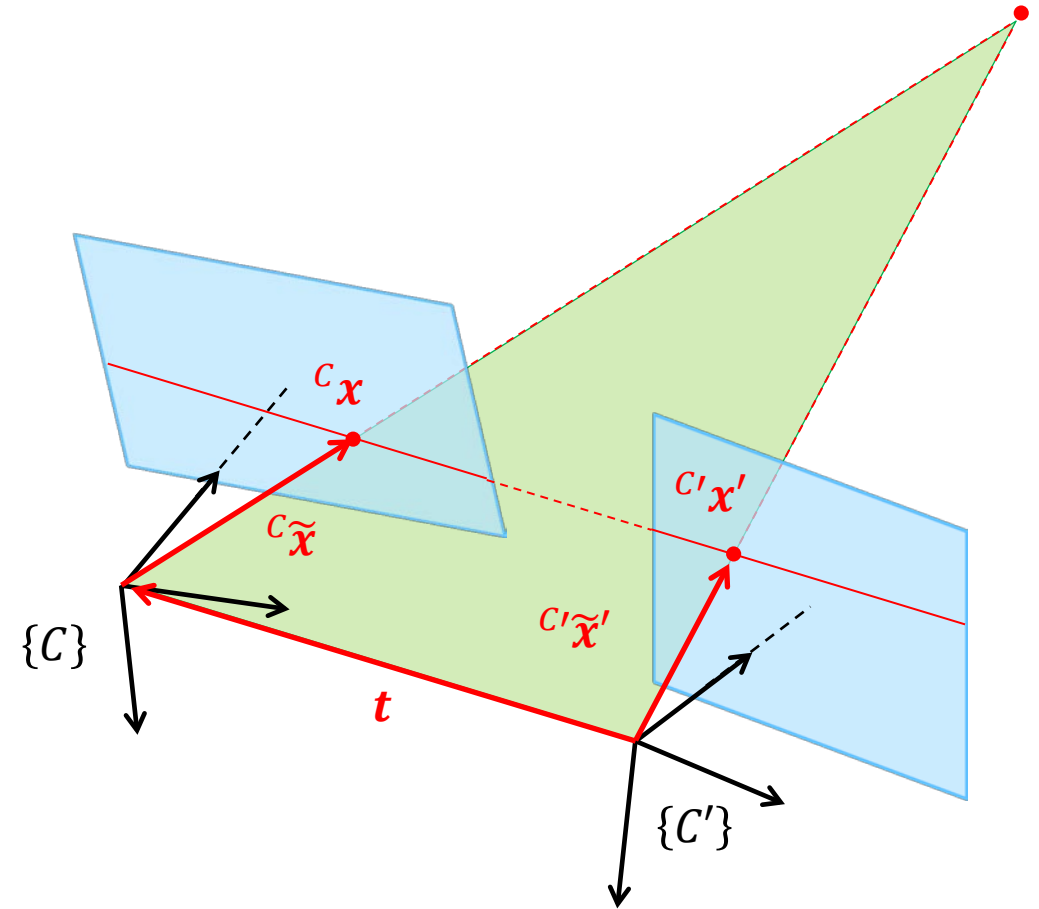
$$({}^{C'}\tilde{\mathbf{x}}' \times \mathbf{t}) \cdot (R {}^C\tilde{\mathbf{x}}) = 0$$

- Rewritten in terms of matrices this takes the form

$${}^{C'}\tilde{\mathbf{x}}'^T [\mathbf{t}]_{\times} R {}^C\tilde{\mathbf{x}} = 0$$

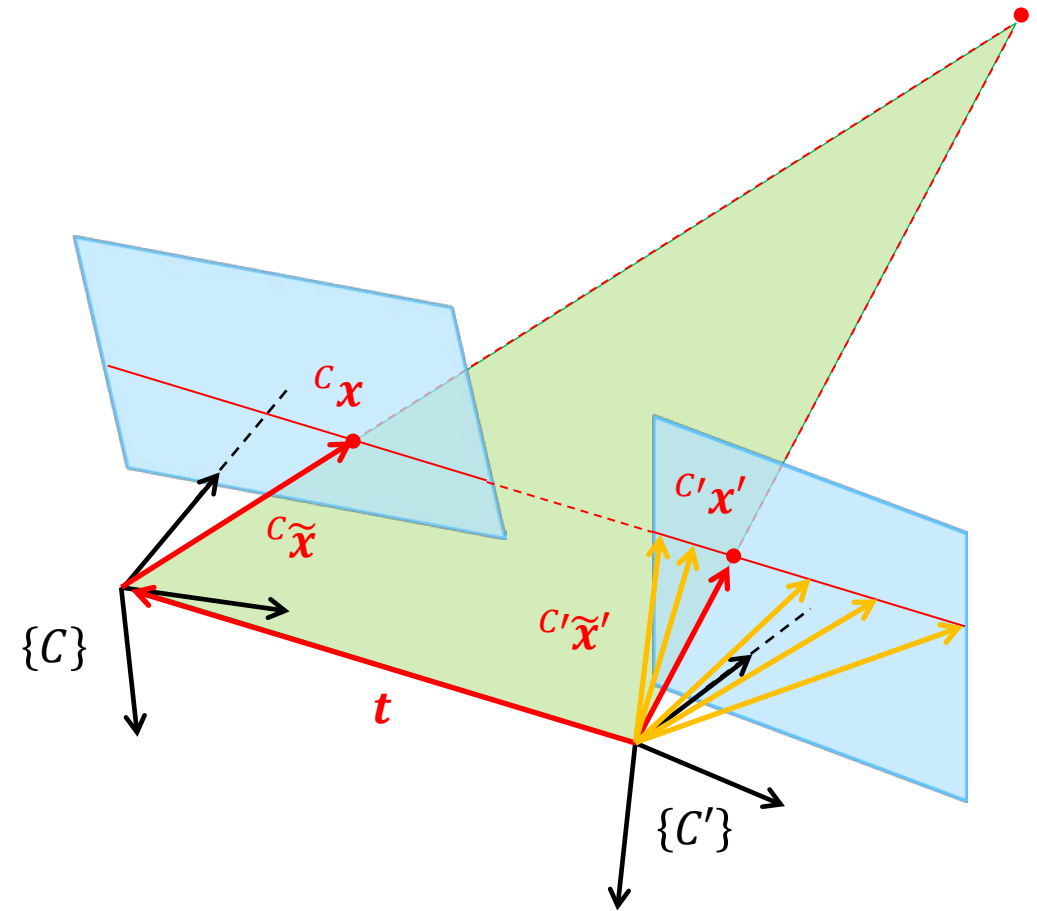
- This relationship defines the essential matrix
 $E = [\mathbf{t}]_{\times} R$

$$\tilde{\mathbf{x}}'^T E \tilde{\mathbf{x}} = 0$$



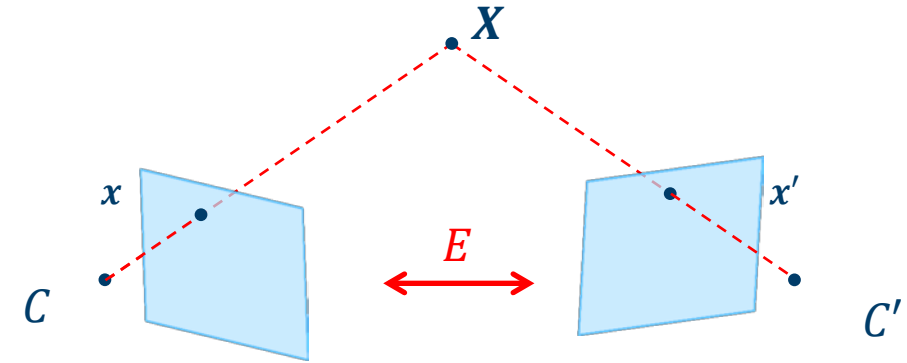
The essential matrix E

- The essential matrix E represents the epipolar constraint on corresponding normalized points
- Note that although $\tilde{x}'^T E \tilde{x} = 0$ is a **necessary** requirement for the correspondence $x \leftrightarrow x'$ to be geometrically possible, it does **not guarantee its correctness**



The essential matrix E

- Properties of E
 - $E = [t]_{\times} R$
 - Homogeneous
 - $rank = 2$
 - $\det = 0$
 - 5 degrees of freedom
 - E can be estimated from a minimum of 5 point correspondences
 - If x and x' are corresponding normalized image points, then $\tilde{x}'^T E \tilde{x} = 0$
 - E has 2 singular values that are equal and a third that is zero



- It is possible to decompose $E = [t]_{\times} R$ to determine the relative pose between cameras
 - Translation only up to scale
 - Topic of another lecture

The fundamental matrix F

- The epipolar constraint on image points is naturally connected to the essential matrix by the calibration matrices K and K'

$$K^C \tilde{x} = \tilde{u} \Rightarrow {}^C \tilde{x} = K^{-1} \tilde{u}$$

$$K'^{C'} \tilde{x}' = \tilde{u}' \Rightarrow {}^{C'} \tilde{x}' = K'^{-1} \tilde{u}' \Rightarrow {}^{C'} \tilde{x}'^T = \tilde{u}'^T K'^{-T}$$

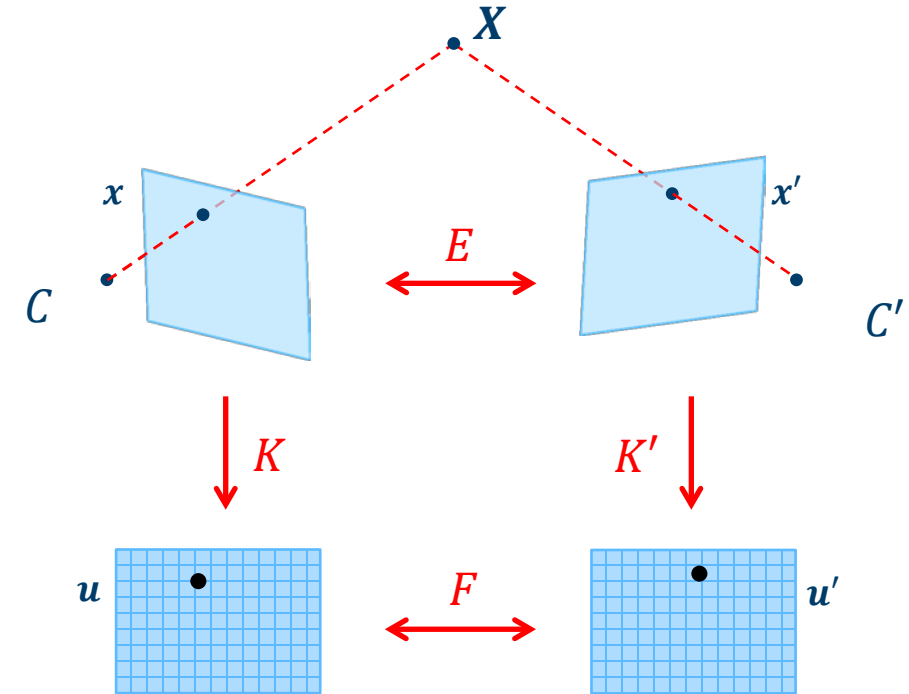
- Combined with the epipolar constraint for normalized image points we get

$${}^{C'} \tilde{x}'^T E {}^C \tilde{x} = 0$$

$$\tilde{u}'^T K'^{-T} E K^{-1} \tilde{u} = 0$$

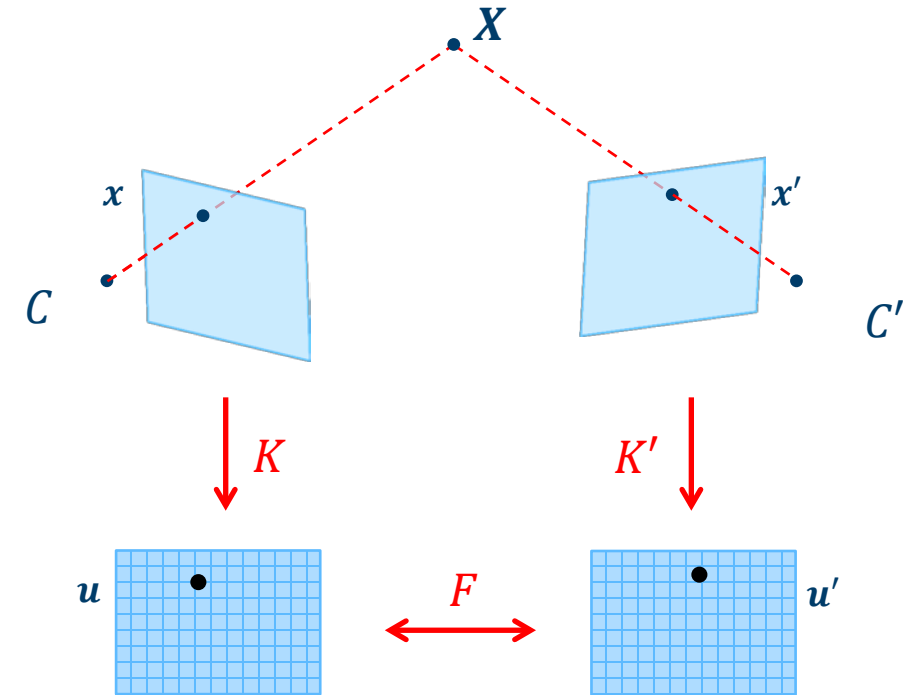
- This defines the fundamental matrix $F = K'^{-T} E K^{-1}$

$$\tilde{u}'^T F \tilde{u} = 0$$



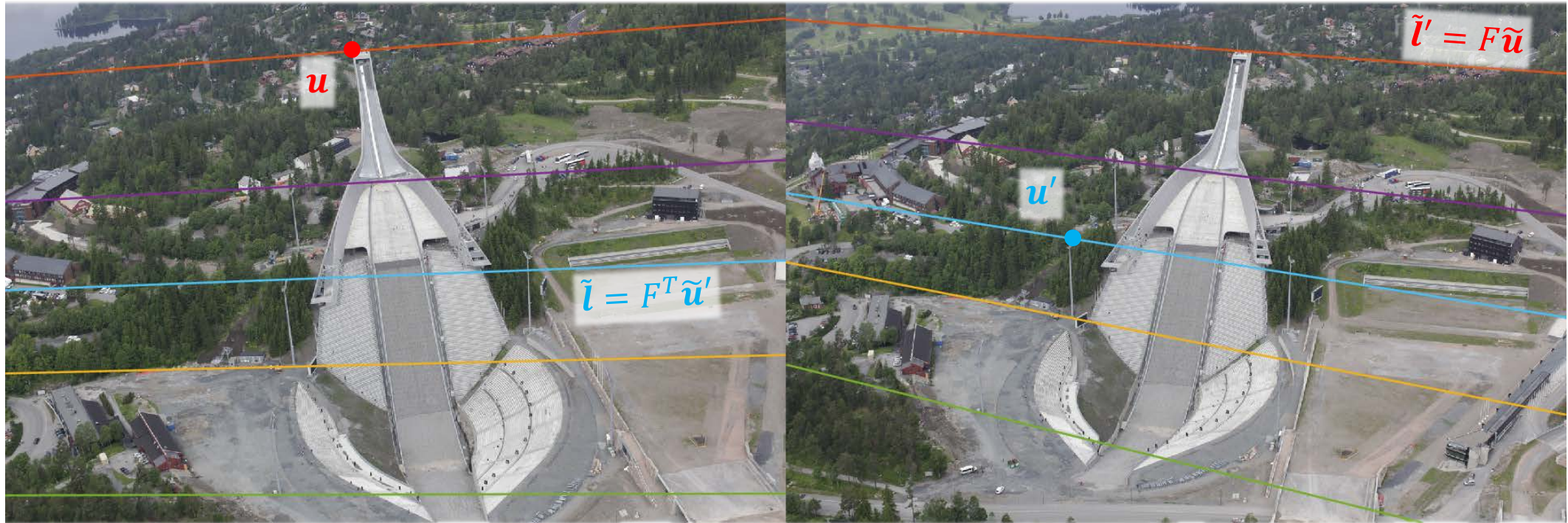
The fundamental matrix F

- Properties of F
 - $F = K'^{-T}EK^{-1}$
 - Homogeneous
 - $\text{rank} = 2$
 - $\det = 0$
 - 7 degrees of freedom
 - F can be estimated from a minimum of 7 point correspondences
 - If $\mathbf{u}$ and $\mathbf{u}'$ are corresponding image points, then $\tilde{\mathbf{u}}'^T F \tilde{\mathbf{u}} = 0$
 - For any point $\mathbf{u}$ in image 1, the corresponding epipolar line $\mathbf{l}'$ in image 2 is given by $\tilde{\mathbf{l}}' = F \tilde{\mathbf{u}}$
 - For any point $\mathbf{u}'$ in image 2, the corresponding epipolar line $\mathbf{l}$ in image 1 is given by $\tilde{\mathbf{l}} = F^T \tilde{\mathbf{u}}'$



- The epipole $\mathbf{e}'$ in image 2 is F 's left singular vector corresponding to the zero singular value
- The epipole $\mathbf{e}$ in image 1 is F 's right singular vector corresponding to the zero singular value

Example



- These fundamental lines were determined using the fundamental matrix between images
- Recall that points and lines are dual in $\mathbb{P}^2$

$$\tilde{l}^T \tilde{u} = 0 \Leftrightarrow \begin{bmatrix} l_0 & l_1 & l_2 \end{bmatrix} \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = 0 \Leftrightarrow l_0 u + l_1 v + l_2 = 0$$

Estimating F

- Several algorithms
 - Non-iterative: 7-pt, 8-pt
 - Iterative: Minimize epipolar error
 - Robust: RANSAC with 7-pt
- From the definition it follows that each point correspondence $\mathbf{u}_i \leftrightarrow \mathbf{u}_i'$ contributes with 1 equation

$$\mathbf{u}_i'^T \mathbf{F} \mathbf{u}_i = 0$$

$$\begin{bmatrix} u_i' & v_i' & 1 \end{bmatrix} \begin{bmatrix} f_1 & f_2 & f_3 \\ f_4 & f_5 & f_6 \\ f_7 & f_8 & f_9 \end{bmatrix} \begin{bmatrix} u_i \\ v_i \\ 1 \end{bmatrix} = 0$$

$$\begin{bmatrix} u_i u_i' & u_i' v_i & u_i' & u_i v_i' & v_i v_i' & v_i' & u_i & v_i & 1 \end{bmatrix} \mathbf{f} = 0$$

- So given several correspondences we get a homogeneous system of linear equations that we can solve by SVD

$$\begin{bmatrix} u_1 u_1' & u_1' v_1 & u_1' & u_1 v_1' & v_1 v_1' & v_1' & u_1 & v_1 & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ u_n u_n' & u_n' v_n & u_n' & u_n v_n' & v_n v_n' & v_n' & u_n & v_n & 1 \end{bmatrix} \mathbf{f} = 0$$

$$A \mathbf{f} = 0$$

- As before, we see that the matrix A contains terms that can be very different in scale, so point sets $\{\mathbf{u}_i\}$ and $\{\mathbf{u}_i'\}$ should be normalized in advance
 - Centroid $\rightarrow$ origin
 - Mean distance from origin should be $\sqrt{2}$

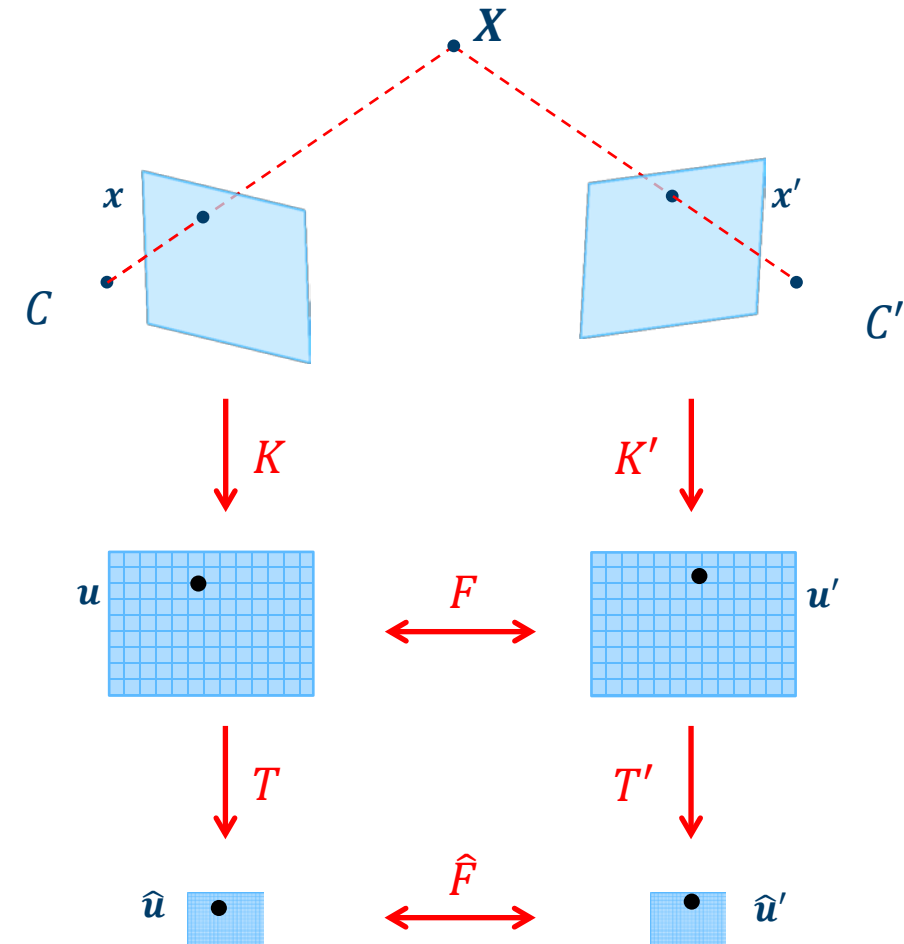
Estimating F

The normalized 8-point algorithm

Given $n \geq 8$ correspondences $\mathbf{u}_i \leftrightarrow \mathbf{u}_i'$, do the following

1. Normalize points $\{\mathbf{u}_i\}$ and $\{\mathbf{u}_i'\}$ using similarity transforms T and T'
2. Build matrix A from point-correspondences and compute its SVD
3. Extract the “solution” $\bar{F}$ from the right singular vector corresponding to the smallest singular value
4. Compute the SVD of $\bar{F}$: $\bar{F} = USV^T$
5. Enforce zero determinant by setting $s_{33} = 0$ and compute a proper fundamental matrix

$$\hat{F} = USV^T$$
6. Denormalize $F = T'^T \hat{F} T$



$$\begin{aligned} \hat{u}'^T \hat{F} \hat{u} &= 0 \\ (T'u')^T \hat{F} (Tu) &= 0 \Rightarrow F = T'^T \hat{F} T \\ u'^T T'^T \hat{F} T u &= 0 \end{aligned}$$

Estimating F

The normalized 8-point algorithm

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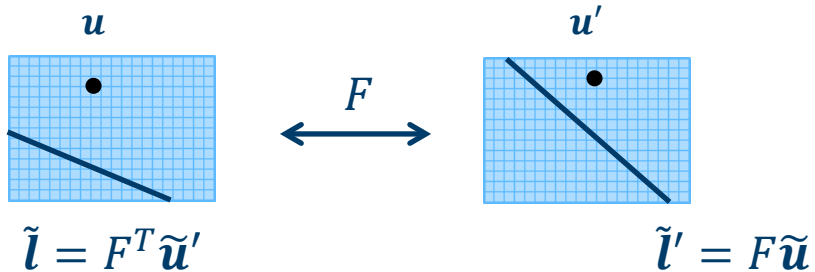
The 7-point algorithm

- Given 7 correspondences, A will be a 7×9 matrix which in general will be of rank 7
- So the null space of A is 2-dimensional and the fundamental matrix must be a linear combination of the two right null vectors of A
$$\mathbf{f}(\alpha) = \alpha \mathbf{f}_1 + (1 - \alpha) \mathbf{f}_2$$
$$F(\alpha) = \alpha F_1 + (1 - \alpha) F_2$$
- The additional constraint $\det(F) = 0$ leads to a cubic polynomial equation in α which has 1 or 3 solutions α_i which in turn yields 1 or 3 F 's
- This algorithm is to prefer in a RANSAC scheme, since it is minimal and since for a single sampling of 7 correspondences one might get 3 fundamental matrices to test for inliers

Estimating F

- Improved estimates of F can be obtained using iterative methods like Levenberg-Marquardt to minimize the epipolar error

$$\epsilon = \sum d(\tilde{\mathbf{u}}, F^T \tilde{\mathbf{u}}') + d(\tilde{\mathbf{u}}', F \tilde{\mathbf{u}})$$

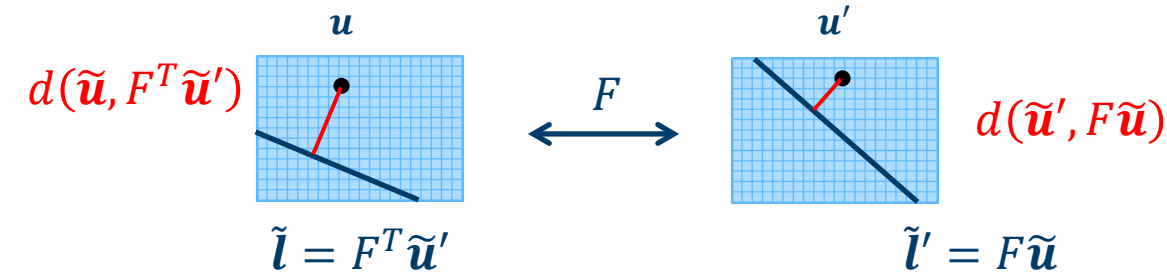


- OpenCV
 - `cv::Mat cv::findFundamentalMat`
 - Arguments are
 - `InputArray` points1
 - `InputArray` points2
 - `int` method – {`CV_FM_7POINT`, `CV_FM_8POINT`, `CV_FM_RANSAC`, `CV_FM_LMEDS`}
 - `double` param1
 - `double` param2
 - `OutputArray` mask
- Matlab
 - `estimateFundamentalMatrix`

Estimating F

- Improved estimates of F can be obtained using iterative methods like Levenberg-Marquardt to minimize the epipolar error

$$\epsilon = \sum d(\tilde{\mathbf{u}}, F^T \tilde{\mathbf{u}}') + d(\tilde{\mathbf{u}}', F \tilde{\mathbf{u}})$$



- Distance between homogeneous point $\mathbf{u}$ and line $\mathbf{l}$, with the homogeneous representation $\tilde{\mathbf{l}} = [\tilde{l}_1, \tilde{l}_2, \tilde{l}_3]^T$

$$d(\tilde{\mathbf{u}}, \tilde{\mathbf{l}}) = \frac{\tilde{\mathbf{u}}^T \tilde{\mathbf{l}}}{\sqrt{\tilde{l}_1^2 + \tilde{l}_2^2}}$$

- OpenCV
 - `cv::Mat cv::findFundamentalMat`
 - Arguments are
 - `InputArray points1`
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 - `int method` – {`CV_FM_7POINT`, `CV_FM_8POINT`, `CV_FM_RANSAC`, `CV_FM_LMEDS`}
 - `double param1`
 - `double param2`
 - `OutputArray mask`
- Matlab
 - `estimateFundamentalMatrix`

Estimating E

- For calibrated cameras (K and K' are known), we can first estimate F and then compute $E = K'^T F K$
- It is also possible to estimate E directly from 5 **normalized** point correspondences $x_i \leftrightarrow x_i'$
 - Algorithm proposed by David Nistér in 2004
 - Involves finding the roots of a 10th degree polynomial
- In **RANSAC schemes**, the 5-point algorithm is the fastest alternative
 - To achieve 99% confidence with 50% outliers, the **5-point** algorithm only requires **145 tests** while the **8-point** algorithm requires **1177 tests**
- OpenCV
 - `cv::Mat cv::findEssentialMat`
 - 5-pt algorithm
- Matlab
 - Currently not available as a built in function in Matlab
 - MexOpenCV
- OpenGV: <http://laurentkneip.github.io/opengv/> contains several interesting functions
 - 5-pt algorithm
 - 2-pt algorithm based on known relative rotation

Summary

- Algebraic representation of epipolar geometry
 - The essential matrix
 - The fundamental matrix
- Estimating the epipolar geometry
 - Estimate F : 7pt, 8pt, RANSAC
 - Estimate E : 5pt
- Additional reading:
 - Szeliski: 7.2

